

MAINTENANCE OPERATIONS CLIENT

Business Requirements Document (BRD)

Maintenance Operations Analytics & Reporting Automation

Document Control	Detail
Engagement Period	Aug–Nov 2024
Role	Business Analyst + Data Analyst
Primary Stakeholder	Maintenance Manager
Data Scope	Approximately 10,000 historical records across 20 teams
Solution	Power BI dashboard built from scratch
Document Status	Portfolio reconstruction based on confirmed engagement facts
Purpose	Define the business requirements, reporting needs, business rules and acceptance expectations for the maintenance analytics solution.

Integrity note: This is a portfolio reconstruction of the BRD, not a claim that this exact document was historically signed or approved by the client. Confirmed facts are separated from reconstructed requirement wording.

1. Business Context

Maintenance reporting depended on manually consolidating operational updates from Excel, CSV, Google Sheets and database sources. The Maintenance Manager needed a consistent way to understand work completion, overdue work, backlog, team performance and technician performance across approximately 10,000 historical records spanning 20 teams.

The source data required cleansing before reliable reporting because it contained duplicates, missing values, inconsistent team names, inconsistent date formats, inconsistent status names and blank records.

2. Business Problem Statement

Problem: Fragmented maintenance data and manual consolidation limited timely, consistent visibility into operational performance and required approximately 3–4 hours of recurring weekly reporting effort.

The reporting solution therefore needed to address both information quality and management visibility rather than simply reproduce existing manual reports.

3. Business Objectives

- Consolidate maintenance information into a structured analytical reporting layer.
- Standardize operational data sufficiently for reliable KPI calculation.
- Provide consistent visibility into completed, open, pending and overdue maintenance work.
- Enable analysis of completion, backlog, timeliness, priority, teams, technicians, maintenance categories and assets/equipment.
- Provide the Maintenance Manager with reusable Power BI reporting.
- Reduce repetitive manual consolidation and recurring reporting effort.
- Support adoption through walkthrough, presentation, documentation and training.

4. Stakeholder & Primary User

Stakeholder	Role in Solution	Primary Need
Maintenance Manager	Primary business stakeholder and dashboard user	Consolidated operational visibility, exception monitoring and performance reporting

5. Scope

In Scope	Description
Historical maintenance data	Approximately 10,000 records across 20 teams
Source consolidation	Excel, CSV, Google Sheets and database data
Data preparation	Cleaning, structuring and standardization
Operational KPI framework	Completion, status, overdue, completion time, priority and performance views
Operational analysis	Backlog, workload imbalance, high-priority pending work, recurring categories and equipment issues
Power BI reporting	Dashboard built from scratch
Delivery/adoption	Live walkthrough, presentation, documentation and training

Not evidenced: Exact historical dashboard page names, original source schema, exact code, formal UAT process, production deployment architecture and exact recommendation wording are not retained.

6. Source Data Requirements

Data Domain	Required / Available Fields
Work identification	Task / Work Order ID
Organization	Team, department
Resource	Technician
Asset	Asset / equipment
Dates	Created Date, Due Date, Completion Date
Workflow	Status
Criticality	Priority
Classification	Category
Operational context	Location, remarks

The reporting process must preserve the business meaning of these fields while standardizing values required for analysis.

7. Data Quality Requirements

ID	Requirement	Business Rationale
DQR-01	Potential duplicate records must be identified and controlled before aggregation.	Prevent inflated task/work-order counts.
DQR-02	Missing values in analytically important fields must be identified and handled transparently.	Avoid silent misclassification.
DQR-03	Team names must be standardized to consistent reporting categories.	Prevent one team appearing as multiple entities.
DQR-04	Date values must be converted to consistent valid date formats.	Support due-date and completion-time logic.
DQR-05	Status values must be standardized sufficiently for completed/open/pending and overdue reporting.	Ensure KPI consistency.
DQR-06	Blank/non-actionable records must be identified and excluded or controlled appropriately.	Prevent meaningless aggregation.
DQR-07	Transformations must not fabricate missing operational facts.	Maintain analytical integrity.

8. Business Requirements

ID	Business Requirement	Priority
BR-01	Provide an overall maintenance operations summary.	High
BR-02	Provide visibility into total assigned maintenance work and completion.	High
BR-03	Provide visibility into completed work orders/tasks.	High
BR-04	Provide visibility into open work orders/tasks.	High
BR-05	Provide visibility into pending maintenance work.	High
BR-06	Identify overdue maintenance work using the agreed overdue rule.	High
BR-07	Provide average completion-time analysis where valid created/completion dates are available.	High
BR-08	Provide maintenance-work breakdown by priority.	High
BR-09	Provide team-wise completion/performance visibility.	High
BR-10	Provide technician-level performance visibility.	High
BR-11	Provide backlog visibility.	High
BR-12	Support identification of workload imbalance.	Medium
BR-13	Provide visibility into high-priority pending work.	High
BR-14	Provide analysis of recurring maintenance categories.	Medium
BR-15	Provide asset/equipment-level breakdown and issue visibility.	Medium
BR-16	Provide interactive Power BI reporting for management use.	High
BR-17	Reduce repetitive weekly manual reporting effort.	High
BR-18	Support management adoption through walkthrough, documentation and training.	Medium

9. Core Business Rules

BRULE-01 — Completion Rate: $\text{Completed Maintenance Tasks} \div \text{Total Assigned Maintenance Tasks} \times 100$

BRULE-02 — Overdue: $\text{Due Date} < \text{Current Date AND Status} \neq \text{Completed}$

These two rules are confirmed from the engagement and must remain consistent throughout the case study.

10. Reconstructed KPI Rule Boundaries

The project also analyzed open work, pending work, average completion time, backlog and other operational measures. Their exact historical implementation logic is not retained. The portfolio may define reproducible calculation logic, but it must be labeled as reconstructed rather than historical.

KPI / Concept	Requirement Boundary
Open Work Orders	Use standardized status logic; exact historical status mapping not claimed.
Pending Tasks	Use documented rebuilt status mapping; do not assume 'open' and 'pending' are identical.
Average Completion Time	Calculate only for records with valid dates and a documented date-difference convention.
Backlog	Define explicitly in the portfolio before calculation; exact historical backlog rule not retained.
Technician Performance	Use multiple operational measures where possible; avoid unsupported single-score ranking.
Workload Imbalance	Use a documented workload comparison method in the rebuild; exact historical threshold not retained.
Recurring Categories	Use frequency/trend analysis without inventing historical frequency thresholds.
Asset/Equipment Issues	Use supported maintenance-record patterns without claiming unsupported failure causation.

11. Reporting Requirements

ID	Requirement
RR-01	Dashboard must present an executive operational summary.
RR-02	Dashboard must support work-order/status analysis.
RR-03	Dashboard must support overdue/backlog exception analysis.
RR-04	Dashboard must support priority analysis.
RR-05	Dashboard must support team-level performance analysis.
RR-06	Dashboard must support technician-level analysis.
RR-07	Dashboard must support maintenance-category analysis.
RR-08	Dashboard must support asset/equipment analysis.
RR-09	Dashboard must provide appropriate interactive filtering.
RR-10	Equivalent KPIs must use consistent definitions across views.

12. Portfolio Dashboard Organization

The exact historical page structure is not retained. For the portfolio reconstruction, the reporting requirements may be organized into the following logical pages:

Logical Page	Purpose
Executive Maintenance Overview	Headline operational KPIs and maintenance status
Work Order & Completion Analysis	Status, completion, timeliness and work-order trends
Team & Technician Performance	Team/technician activity and performance visibility
Backlog, Priority & Asset Analysis	Exceptions, high-priority work, recurring categories and asset/equipment patterns

Classification: These page names are proposed portfolio architecture, not exact historical dashboard page names.

13. Functional Requirements

ID	Functional Requirement
FR-01	Load structured maintenance data into the analytical reporting model.
FR-02	Apply documented standardization to team, status and date fields.
FR-03	Calculate completion rate using the confirmed business rule.
FR-04	Identify overdue work using the confirmed due-date/status rule.
FR-05	Aggregate operational measures by team, technician, priority, category and asset/equipment as applicable.
FR-06	Allow business users to filter relevant dashboard views.
FR-07	Maintain KPI consistency when filters are applied.
FR-08	Display exceptions such as overdue and high-priority pending work clearly.
FR-09	Support recurring management reporting through the Power BI dashboard.
FR-10	Provide user-facing documentation/training material sufficient to explain the reporting solution.

14. Non-Functional / Quality Requirements

ID	Requirement
NFR-01	Reporting calculations must be reproducible from prepared source data.
NFR-02	Data-quality exceptions must not be silently converted into valid business facts.
NFR-03	Dashboard terminology must be consistent and understandable to the Maintenance Manager.
NFR-04	Dashboard interactions must not create inconsistent KPI values across views.
NFR-05	The public portfolio version must not expose confidential client, employee or operationally sensitive data.
NFR-06	Unknown historical details must be labeled as reconstructed/proposed rather than represented as original facts.
NFR-07	The reporting solution should reduce repetitive manual reporting activity compared with the prior process.

15. Data Transformation Requirements

Transformation Area	Requirement
Duplicates	Detect and resolve/control duplicates using evidence-based logic.
Missing values	Flag, retain, exclude or remediate according to analytical impact; do not invent values.
Team names	Map inconsistent labels to a controlled team representation.
Dates	Standardize date types and validate Created/Due/Completion chronology where relevant.
Statuses	Map inconsistent status labels to documented analytical categories.
Blank records	Prevent empty/non-actionable records from distorting KPI calculations.
Identifiers	Preserve work-order/task uniqueness sufficiently for reliable aggregation.

16. Analytical Requirements

ID	Analysis Requirement
AR-01	Determine overall completion rate.
AR-02	Analyze completed, open and pending work.
AR-03	Identify overdue work.
AR-04	Analyze average completion time.
AR-05	Compare performance across teams.
AR-06	Analyze technician performance.
AR-07	Analyze maintenance work by priority.
AR-08	Identify backlog.
AR-09	Identify workload imbalance.
AR-10	Identify high-priority pending work.
AR-11	Identify recurring maintenance categories.
AR-12	Analyze asset/equipment-related patterns or issues.

17. Acceptance Criteria

AC ID	Acceptance Criterion
AC-01	Prepared data supports reliable reporting across approximately 10,000 historical records and 20 teams, subject to documented data-quality handling.
AC-02	Completion Rate uses Completed Maintenance Tasks ÷ Total Assigned Maintenance Tasks × 100.
AC-03	Overdue logic uses Due Date < Current Date AND Status ≠ Completed.
AC-04	Dashboard supports the confirmed work-order, priority, team, technician, category and asset/equipment analytical areas.
AC-05	Equivalent KPIs reconcile across dashboard views under equivalent filters.
AC-06	Data-quality exceptions do not silently inflate or misclassify operational measures.
AC-07	Management can consume the solution through Power BI and supporting walkthrough/documentation/training.
AC-08	The reporting process reduces repetitive manual reporting effort; the confirmed engagement outcome was approximately 3–4 hours per week eliminated.
AC-09	The 92% completion rate is presented as a dashboard finding, not as a project-generated improvement.
AC-10	Portfolio outputs preserve client confidentiality and clearly label reconstructed implementation details.

18. Requirements Traceability

Business Need	Requirement IDs	Planned Evidence
Completion visibility	BR-02, BR-03, FR-03, AR-01	KPI Catalogue, dashboard, testing
Status visibility	BR-03 to BR-05, AR-02	Dashboard and KPI validation
Overdue visibility	BR-06, FR-04, AR-03	Overdue rule, dashboard, test cases
Timeliness	BR-07, AR-04	KPI definition and validation
Team / technician performance	BR-09, BR-10, AR-05, AR-06	Performance analysis and dashboard
Priority / exceptions	BR-08, BR-13, AR-07, AR-10	Priority/backlog dashboard analysis
Backlog / workload	BR-11, BR-12, AR-08, AR-09	Operational analysis
Category / asset analysis	BR-14, BR-15, AR-11, AR-12	Operational analysis and dashboard
Reporting automation	BR-16, BR-17, FR-09	Power BI solution and confirmed time saving
Adoption	BR-18, FR-10	Walkthrough, documentation and training

19. Constraints & Assumptions

Type	Statement
Constraint	Source data required cleansing and standardization before reliable reporting.
Constraint	Exact historical status mappings, backlog definition and detailed calculation implementation are not retained.
Constraint	Exact original dashboard layout and code are not retained.
Constraint	Client identity and sensitive operational details must remain confidential.
Assumption for reconstruction	Portfolio technical logic may be rebuilt where necessary if clearly labeled as reconstructed.
Assumption for reconstruction	Portfolio dashboard pages may be reorganized for clarity without claiming historical equivalence.

20. Risks & Controls

Risk	Control
Duplicate work orders inflate counts	Duplicate identification and reconciliation
Inconsistent statuses distort KPIs	Controlled status standardization and test cases
Invalid dates distort overdue/completion time	Date parsing and chronology validation
Missing team/technician values affect attribution	Explicit missing-value handling
Reconstructed logic is mistaken for historical logic	Clear reconstruction labels
Confidential operational data appears publicly	Sanitized portfolio dataset and output review
92% is misrepresented as improvement	Explicitly label it as an observed completion rate

21. Business Outcome Reference

Observed KPI: 92% overall completion rate surfaced by the dashboard.

Efficiency outcome: Approximately 3–4 hours of weekly manual reporting effort eliminated.

The BRD does not claim that the project increased completion to 92%, nor does it invent additional operational improvement percentages.

22. Approval / Sign-Off Boundary

A live walkthrough, presentation, documentation and training are confirmed. Formal historical BRD sign-off, approval dates, named approvers or signed UAT acceptance are not retained.

Portfolio status: No fabricated signature block or historical approval is included.

23. BRD Summary

Requirement chain: Fragmented multi-source maintenance reporting → controlled data preparation → consistent maintenance business rules → operational KPI analysis → Power BI management reporting → reduced recurring reporting effort.